In [1]:
```python
import pandas as  pd
```

In [2]:
```python
pd.__version__
```

Out[2]: '2.3.3'

In [3]:
```python
df = pd.read_csv(r"C:\Users\3star\Downloads\data1.csv")
```

In [4]:
```python
df
```

Out[4]:

	Country Name	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [5]:
```python
type(df)
```

Out[5]: pandas.core.frame.DataFrame

In [6]:
```python
len(df)
```

Out[6]: 195

In [7]:
```python
df.columns
```

Out[7]:
```
Index(['Country Name', 'CountryCode', 'BirthRate', 'InternetUsers',
       'IncomeGroup'],
      dtype='object')
```

In [8]:
```python
len(df.columns)
```

Out[8]: 5

In [9]:
```
df.shape      # dimension of table
```
Out[9]:
```
(195, 5)
```
In [10]:
```
df.isnull()
```
Out[10]:

	Country Name	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
190	False	False	False	False	False
191	False	False	False	False	False
192	False	False	False	False	False
193	False	False	False	False	False
194	False	False	False	False	False

195 rows × 5 columns

In [11]:
```
df.isnull().sum()
```
Out[11]:
```
Country Name     0
CountryCode      0
BirthRate        0
InternetUsers    0
IncomeGroup      0
dtype: int64
```
In [12]:
```
df.head()     #bydefault bit print top 5 rows
```
Out[12]:

	Country Name	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

In [13]:
```
df.dtypes
```

Out[13]:
```
Country Name      object
CountryCode       object
BirthRate        float64
InternetUsers    float64
IncomeGroup       object
dtype: object
```

In [14]:
```
df.columns
```

Out[14]:
```
Index(['Country Name', 'CountryCode', 'BirthRate', 'InternetUsers',
       'IncomeGroup'],
      dtype='object')
```

In [15]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 195 entries, 0 to 194
Data columns (total 5 columns):
 #   Column         Non-Null Count  Dtype  
---  ------         --------------  -----  
 0   Country Name   195 non-null    object 
 1   CountryCode    195 non-null    object 
 2   BirthRate      195 non-null    float64
 3   InternetUsers  195 non-null    float64
 4   IncomeGroup    195 non-null    object 
dtypes: float64(2), object(3)
memory usage: 7.7+ KB
```

Slicing in Pandas

In [16]:
```
df[:]
```

Out[16]:

	Country Name	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [17]:

```
df[:6]
```

Out[17]:

	Country Name	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
5	Argentina	ARG	17.716	59.9	High income

In [18]:

```
df[4:]
```

Out[18]:

	Country Name	CountryCode	BirthRate	InternetUsers	IncomeGroup
4	United Arab Emirates	ARE	11.044	88.0	High income
5	Argentina	ARG	17.716	59.9	High income
6	Armenia	ARM	13.308	41.9	Lower middle income
7	Antigua and Barbuda	ATG	16.447	63.4	High income
8	Australia	AUS	13.200	83.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

191 rows × 5 columns

In [19]:

```
df[0:50:5]
```

Out[19]:

	Country Name	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.90	High income
5	Argentina	ARG	17.716	59.90	High income
10	Azerbaijan	AZE	18.300	58.70	Upper middle income
15	Bangladesh	BGD	20.142	6.63	Lower middle income
20	Belarus	BLR	12.500	54.17	Upper middle income
25	Barbados	BRB	12.188	73.00	High income
30	Canada	CAN	10.900	85.80	High income
35	Cameroon	CMR	37.236	6.40	Lower middle income
40	Costa Rica	CRI	15.022	45.96	Upper middle income
45	Germany	DEU	8.500	84.17	High income

In [20]:

```
df[::-10]
```

Out[20]:

	Country Name	CountryCode	BirthRate	InternetUsers	IncomeGroup
194	Zimbabwe	ZWE	35.715	18.5000	Low income
184	Venezuela, RB	VEN	19.842	54.9000	High income
174	Trinidad and Tobago	TTO	14.590	63.8000	High income
164	Swaziland	SWZ	30.093	24.7000	Lower middle income
154	Sierra Leone	SLE	36.729	1.7000	Low income
144	French Polynesia	PYF	16.393	56.8000	High income
134	Oman	OMN	20.419	66.4500	High income
124	Malaysia	MYS	16.805	66.9700	Upper middle income
114	Macedonia, FYR	MKD	11.222	65.2400	Upper middle income
104	Lesotho	LSO	28.738	5.0000	Lower middle income
94	Kiribati	KIR	29.044	11.5000	Lower middle income
84	Iceland	ISL	13.400	96.5468	High income
74	Hong Kong SAR, China	HKG	7.900	74.2000	High income
64	Guinea	GIN	37.337	1.6000	Low income
54	Estonia	EST	10.300	79.4000	High income
44	Czech Republic	CZE	10.200	74.1104	High income
34	Cote d'Ivoire	CIV	37.320	8.4000	Lower middle income
24	Brazil	BRA	14.931	51.0400	Upper middle income
14	Burkina Faso	BFA	40.551	9.1000	Low income
4	United Arab Emirates	ARE	11.044	88.0000	High income

In [21]:

```
df[::-1]
```

Out[21]:

	Country Name	CountryCode	BirthRate	InternetUsers	IncomeGroup
194	Zimbabwe	ZWE	35.715	18.5	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
191	South Africa	ZAF	20.850	46.5	Upper middle income
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
...	...	...	...	...	...
4	United Arab Emirates	ARE	11.044	88.0	High income
3	Albania	ALB	12.877	57.2	Upper middle income
2	Angola	AGO	45.985	19.1	Upper middle income
1	Afghanistan	AFG	35.253	5.9	Low income
0	Aruba	ABW	10.244	78.9	High income

195 rows × 5 columns

In [22]:

```
df.head()
```

Out[22]:

	Country Name	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

In [23]:

```
df.columns
```

Out[23]:

```
Index(['Country Name', 'CountryCode', 'BirthRate', 'InternetUsers',
       'IncomeGroup'],
      dtype='object')
```

In [24]:

```
df.describe()   #It will only read numerical attributes
```

Out[24]:

	BirthRate	InternetUsers
count	195.000000	195.000000
mean	21.469928	42.076471
std	10.605467	29.030788
min	7.900000	0.900000
25%	12.120500	14.520000
50%	19.680000	41.000000
75%	29.759500	66.225000
max	49.661000	96.546800

In [25]:

```
#If i want to split the data

df.head()
```

Out[25]:

	Country Name	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

In [26]:

```
df[['Country Name', 'CountryCode', 'IncomeGroup']]
```

Out[26]:

	Country Name	CountryCode	IncomeGroup
0	Aruba	ABW	High income
1	Afghanistan	AFG	Low income
2	Angola	AGO	Upper middle income
3	Albania	ALB	Upper middle income
4	United Arab Emirates	ARE	High income
...	...	...	...
190	Yemen, Rep.	YEM	Lower middle income
191	South Africa	ZAF	Upper middle income
192	Congo, Dem. Rep.	COD	Low income
193	Zambia	ZMB	Lower middle income
194	Zimbabwe	ZWE	Low income

195 rows × 3 columns

In [27]:

```
df_rat = df[['Country Name', 'CountryCode', 'IncomeGroup']]
df_rat.head(4)
```

Out[27]:

	Country Name	CountryCode	IncomeGroup
0	Aruba	ABW	High income
1	Afghanistan	AFG	Low income
2	Angola	AGO	Upper middle income
3	Albania	ALB	Upper middle income

In [28]:

```
df_num = df[['BirthRate','InternetUsers']]
df_num
```

Out[28]:

	BirthRate	InternetUsers
0	10.244	78.9
1	35.253	5.9
2	45.985	19.1
3	12.877	57.2
4	11.044	88.0
...	...	...
190	32.947	20.0
191	20.850	46.5
192	42.394	2.2
193	40.471	15.4
194	35.715	18.5

195 rows × 2 columns

In [29]:
```
print(df.shape)
print(df_num.shape)
print(df_rat.shape)
```

```
(195, 5)
(195, 2)
(195, 3)
```

In [30]:
```
print(df.columns)
print(df_num.columns)
print(df_rat.columns)
```

```
Index(['Country Name', 'CountryCode', 'BirthRate', 'InternetUsers',
       'IncomeGroup'],
      dtype='object')
Index(['BirthRate', 'InternetUsers'], dtype='object')
Index(['Country Name', 'CountryCode', 'IncomeGroup'], dtype='object')
```

In [31]:
```
df_rat.describe()
```

Out[31]:

	Country Name	CountryCode	IncomeGroup
count	195	195	195
unique	195	195	4
top	Aruba	ABW	High income
freq	1	1	67

In [32]:
```
df_rat.describe().T
```

Out[32]:

	count	unique	top	freq
Country Name	195	195	Aruba	1
CountryCode	195	195	ABW	1
IncomeGroup	195	4	High income	67

In [33]:
```
df_num.describe().T
```

Out[33]:

	count	mean	std	min	25%	50%	75%	max
BirthRate	195.0	21.469928	10.605467	7.9	12.1205	19.68	29.7595	49.6610
InternetUsers	195.0	42.076471	29.030788	0.9	14.5200	41.00	66.2250	96.5468

In [34]:
```
df.head(2)
```

Out[34]:

	Country Name	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income

In [35]:
```
df.columns = ['a', 'b', 'c', 'd', 'e']
```

In [36]:
```
df.head(3)
```

Out[36]:

	a	b	c	d	e
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income

In [37]:
```
df.columns = ['Country Name', 'CountryCode', 'BirthRate','InternetUsers', 'IncomeGr
```

In [38]:
```
df.head(2)
```

Out[38]:

	Country Name	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income

In [39]:
```
#Conditions in pandas

df['InternetUsers'] < 2
```

Out[39]:
```
0      False
1      False
2      False
3      False
4      False
       ...
190    False
191    False
192    False
193    False
194    False
Name: InternetUsers, Length: 195, dtype: bool
```

In [40]:
```
df[df['InternetUsers'] < 2]
```

Out[40]:

	Country Name	CountryCode	BirthRate	InternetUsers	IncomeGroup
11	Burundi	BDI	44.151	1.3	Low income
52	Eritrea	ERI	34.800	0.9	Low income
55	Ethiopia	ETH	32.925	1.9	Low income
64	Guinea	GIN	37.337	1.6	Low income
117	Myanmar	MMR	18.119	1.6	Lower middle income
127	Niger	NER	49.661	1.7	Low income
154	Sierra Leone	SLE	36.729	1.7	Low income
156	Somalia	SOM	43.891	1.5	Low income
172	Timor-Leste	TLS	35.755	1.1	Lower middle income

In [41]:
```
len(df[df['InternetUsers'] < 2])
```

Out[41]:
```
9
```

In [42]:
```
df['BirthRate'] > 40
```

Out[42]:
```
0      False
1      False
2       True
3      False
4      False
       ...
190    False
191    False
192     True
193     True
194    False
Name: BirthRate, Length: 195, dtype: bool
```

In [43]:
```
df[df['BirthRate'] > 40]
```

Out[43]:

	Country Name	CountryCode	BirthRate	InternetUsers	IncomeGroup
2	Angola	AGO	45.985	19.1	Upper middle income
11	Burundi	BDI	44.151	1.3	Low income
14	Burkina Faso	BFA	40.551	9.1	Low income
65	Gambia, The	GMB	42.525	14.0	Low income
115	Mali	MLI	44.138	3.5	Low income
127	Niger	NER	49.661	1.7	Low income
128	Nigeria	NGA	40.045	38.0	Lower middle income
156	Somalia	SOM	43.891	1.5	Low income
167	Chad	TCD	45.745	2.3	Low income
178	Uganda	UGA	43.474	16.2	Low income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income

In [44]:
```
len(df[df['BirthRate'] > 40])
```

Out[44]: 12

In [45]:
```
df[(df.InternetUsers < 2) & (df.BirthRate > 40)]
```

Out[45]:

	Country Name	CountryCode	BirthRate	InternetUsers	IncomeGroup
11	Burundi	BDI	44.151	1.3	Low income
127	Niger	NER	49.661	1.7	Low income
156	Somalia	SOM	43.891	1.5	Low income

In [46]:
```
df['IncomeGroup'].unique()
```

Out[46]:
```
array(['High income', 'Low income', 'Upper middle income',
       'Lower middle income'], dtype=object)
```

In [47]:
```
df['IncomeGroup'].nunique()
```

Out[47]: 4

Visualization

In [48]:
```
import matplotlib.pyplot as plt
import seaborn as sns
```

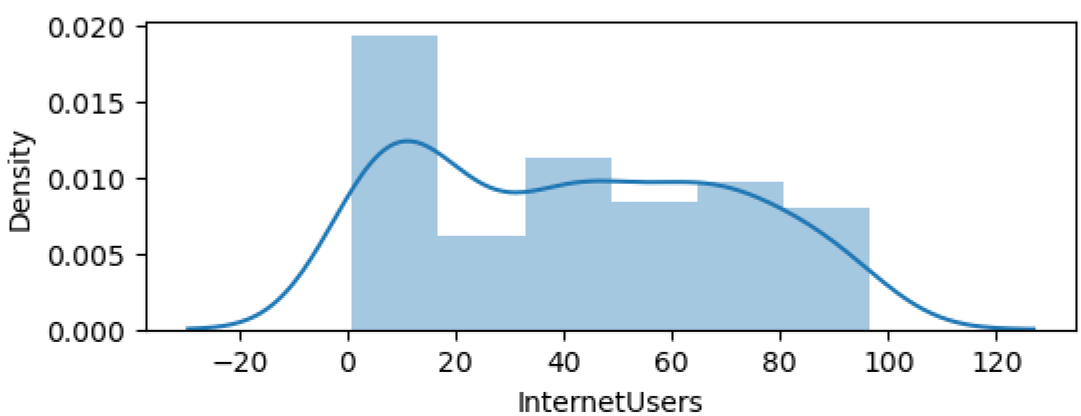
In [49]:
```
%matplotlib inline
plt.rcParams['figure.figsize'] = 6,2
```

In [50]:
```
import warnings
warnings.filterwarnings('ignore')
```

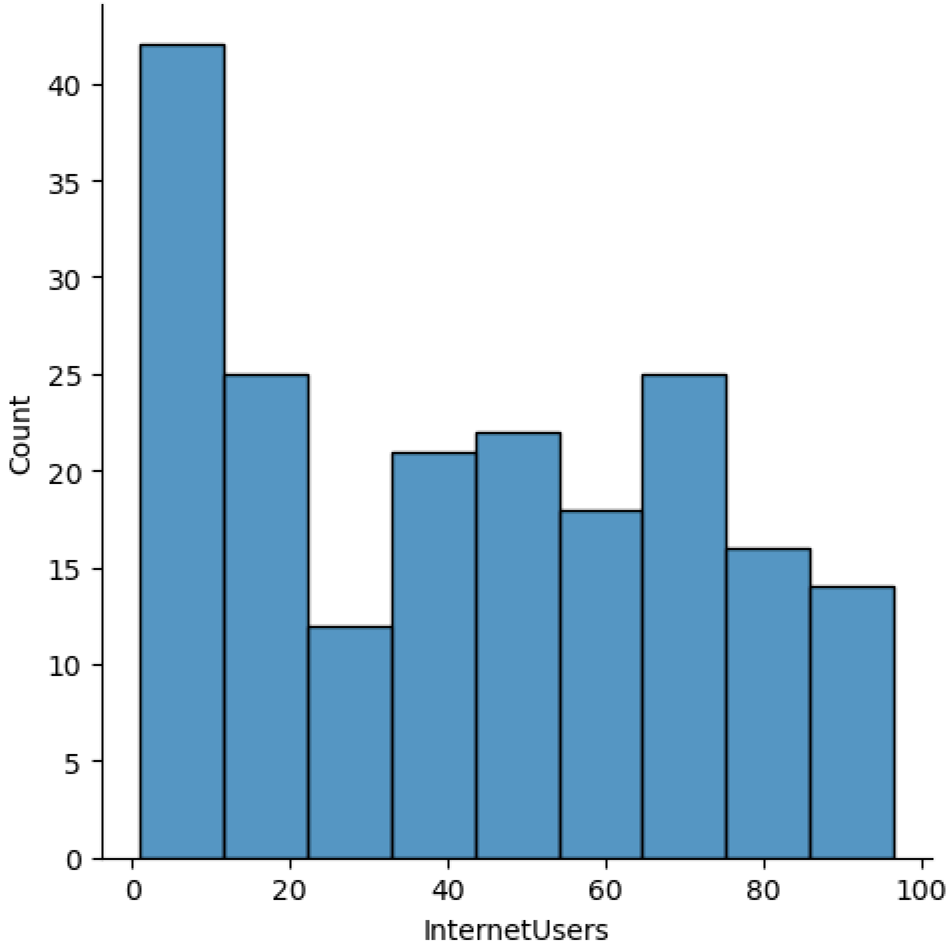
In [51]:
```
df['InternetUsers']
```

Out[51]:
```
0      78.9
1       5.9
2      19.1
3      57.2
4      88.0
       ...
190    20.0
191    46.5
192     2.2
193    15.4
194    18.5
Name: InternetUsers, Length: 195, dtype: float64
```

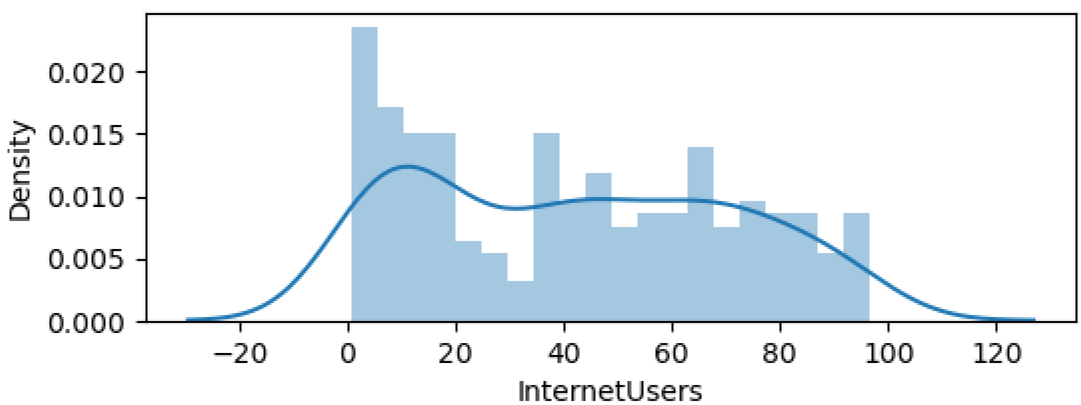
In [52]:
```
vis1 = sns.distplot(df['InternetUsers'])
```



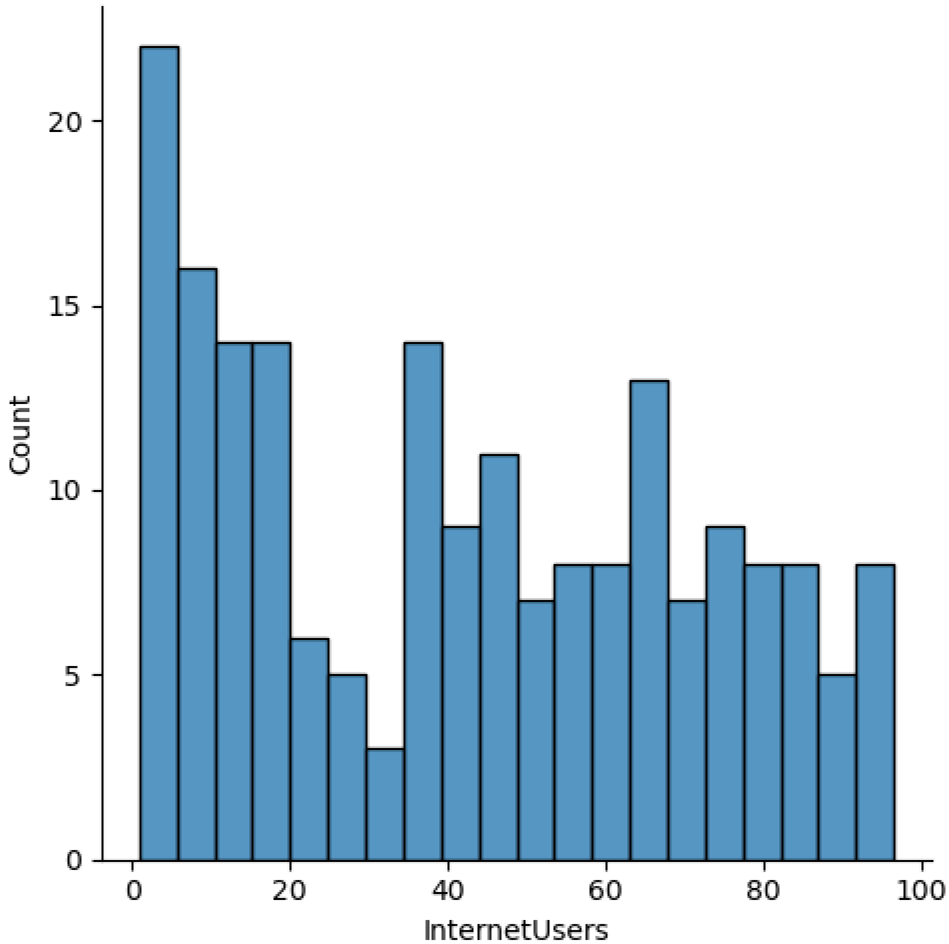
In [53]:
```
vis1 = sns.displot(df['InternetUsers'])
```



In [59]:
```python
vis2 = sns.distplot(df['InternetUsers'], bins = 20)
```

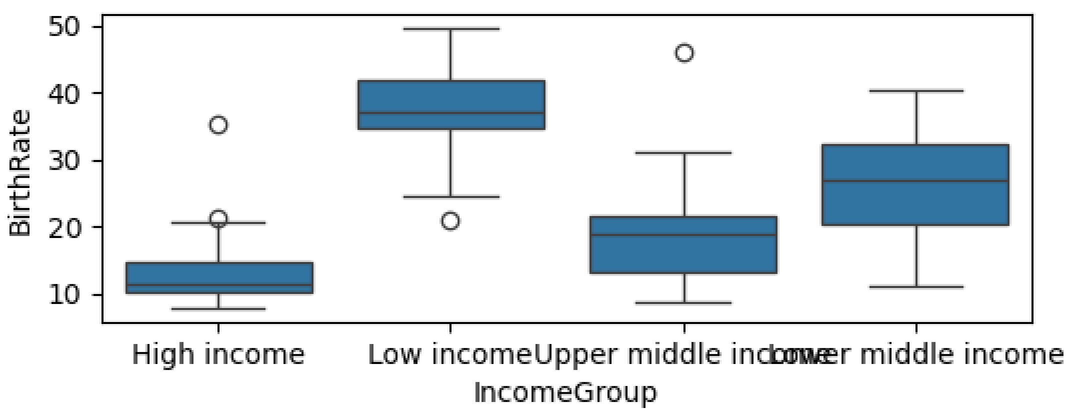


In [60]:
```python
vis2 = sns.displot(df['InternetUsers'], bins = 20)
```



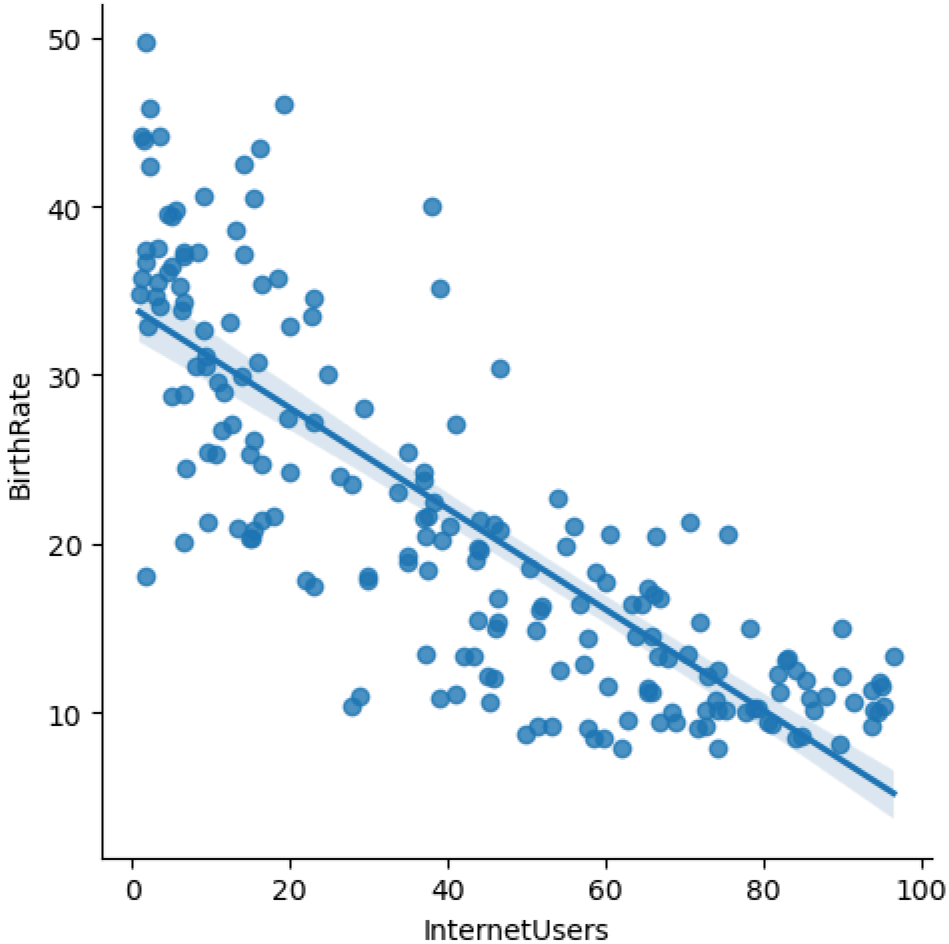
In [62]:
```
# Box Plot

vis4 = sns.boxplot(data = df, x = "IncomeGroup", y = 'BirthRate')
```

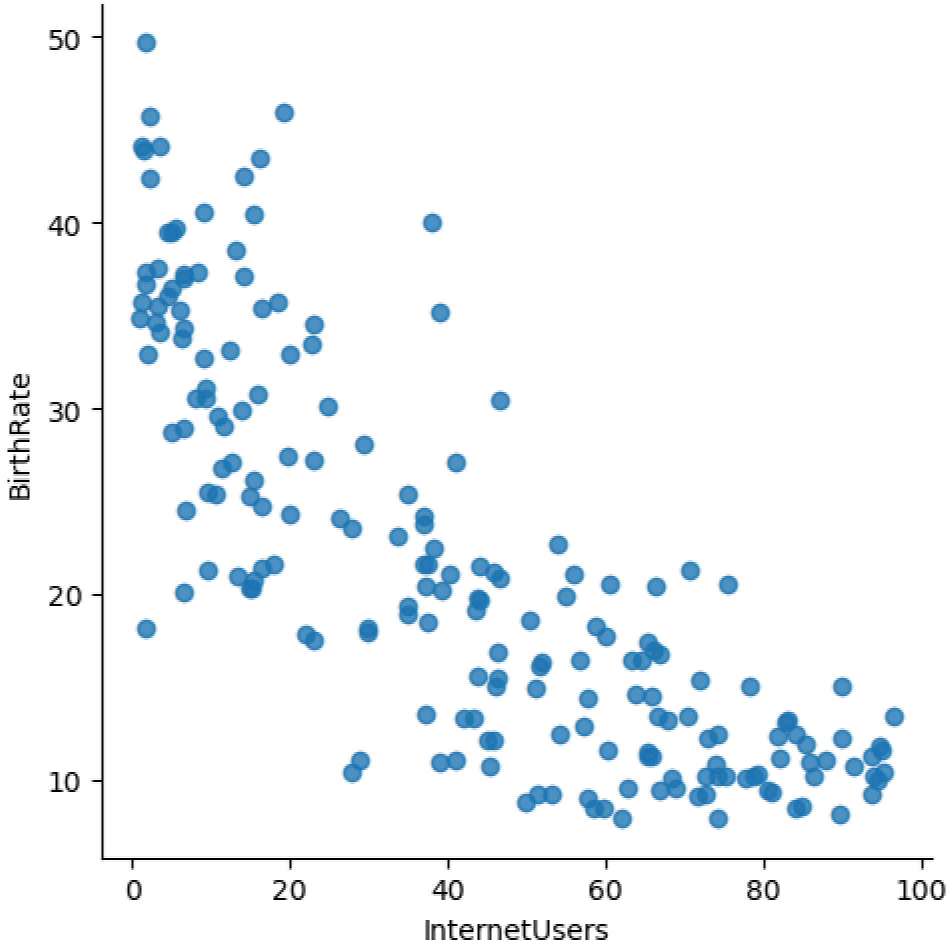


Visualization with seaborn

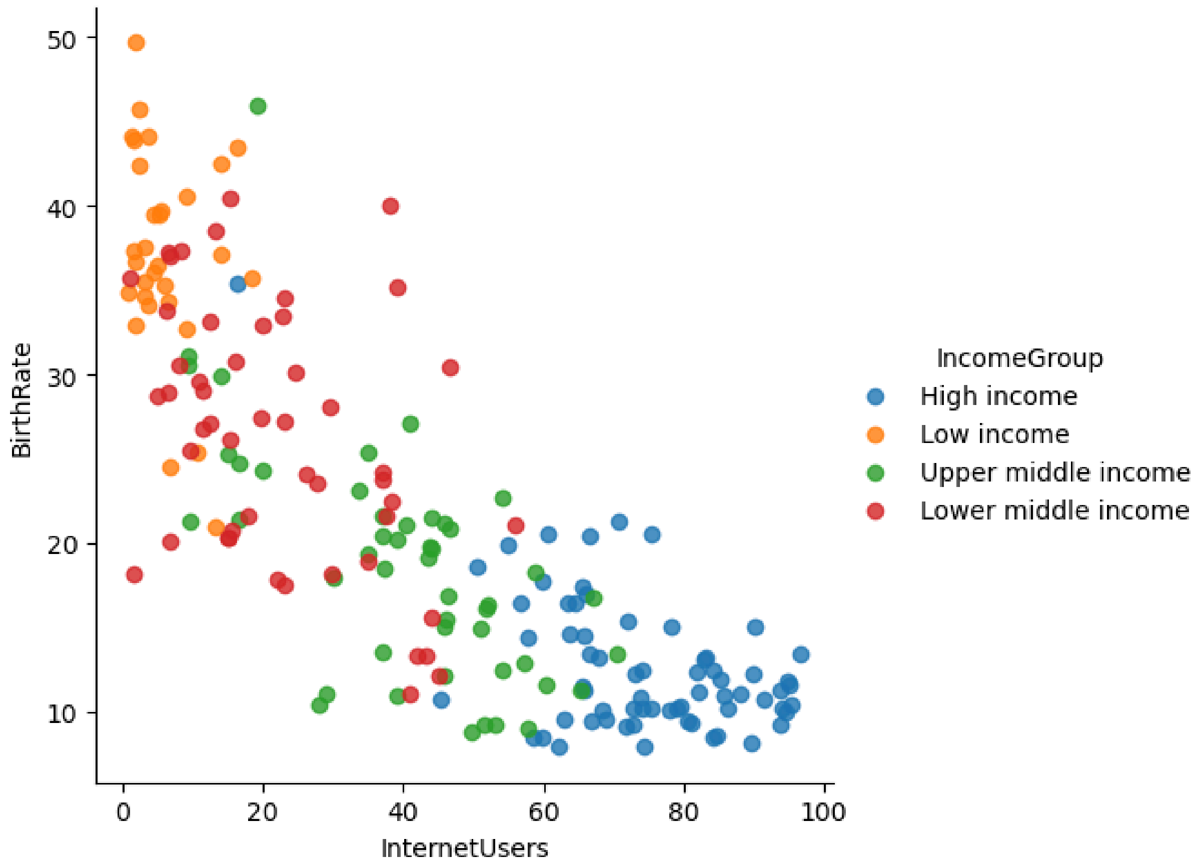
In [65]:
```
vis5 = sns.lmplot(data = df, x = 'InternetUsers', y = 'BirthRate')    #lm = linear m
```



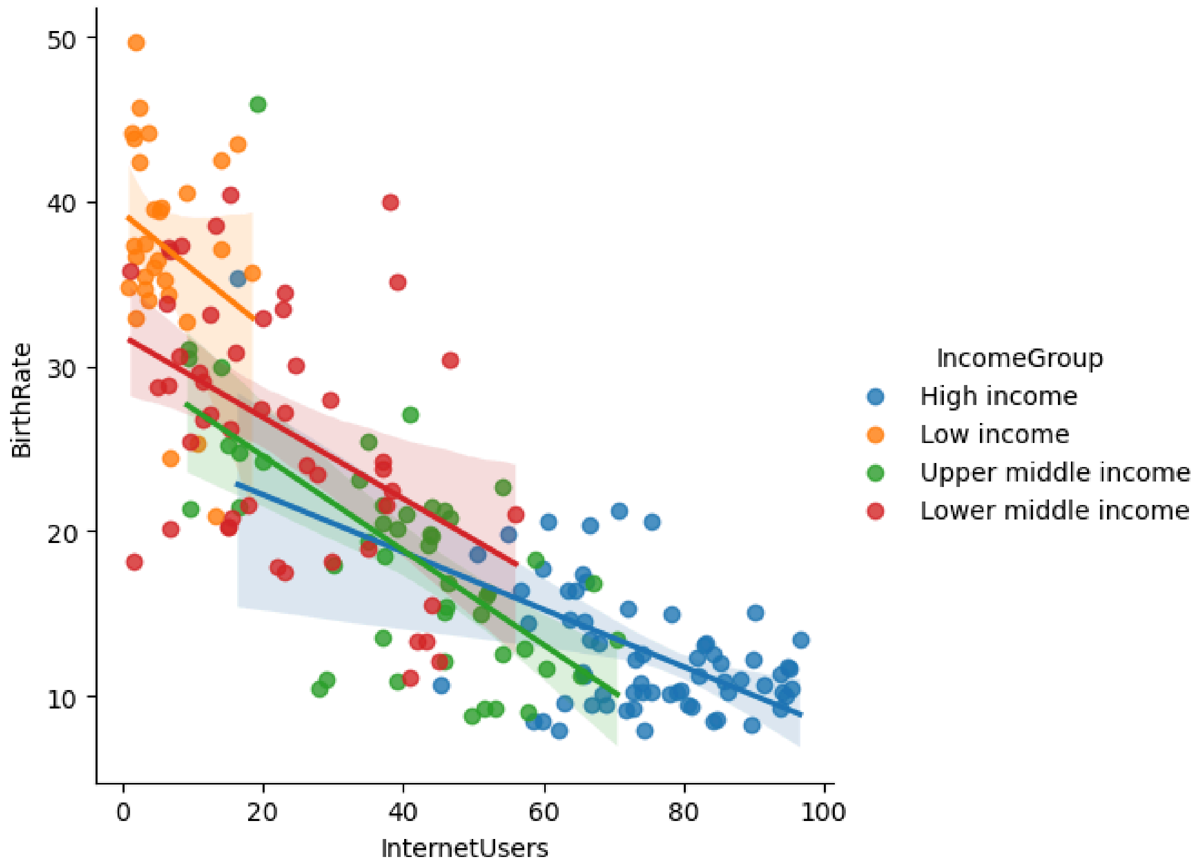
In [69]:
```
vis5 = sns.lmplot(data = df, x = 'InternetUsers', y = 'BirthRate', fit_reg = False)
```




```
In [72]: vis6 = sns.lmplot(data = df, x = 'InternetUsers', y = 'BirthRate', fit_reg = False,
         # hue= parameter for color
```



In [73]:
```
vis7 = sns.lmplot(data = df, x = 'InternetUsers', y = 'BirthRate', fit_reg = True,
# hue= parameter for color
```



BirthRate
InternetUsers
IncomeGroup
High income
Low income
Upper middle income
Lower middle income
50
40
30
20
10
0
20
40
60
80
100

In [ ]: